

Straight lines - Set 1

TMUA practice questions by topic. Specification MM3.1, M4.9, M4.10. 12 questions.

Each question has one correct answer. No calculator. Answers and worked solutions are in the separate solutions PDF.

- 1 Find the equation of the line through $(2, -3)$ that is perpendicular to $3x - 2y + 5 = 0$.

A $2x - 3y - 13 = 0$
B $3x + 2y = 0$
C $2x + 3y + 5 = 0$
D $2x + 3y - 13 = 0$
E $2x + 3y - 5 = 0$
F $3x - 2y - 12 = 0$

- 2 The points $A(1, 2)$, $B(5, 4)$ and $C(k, 8)$ are collinear. Find k .

A 12
B 11
C 10
D 9
E 7
F 13

- 3 Which of the following lines is parallel to $4x + 6y = 9$?

A $2x + 3y = 1$
B $3x - 2y = 4$
C $y = \frac{3}{2}x + 1$
D $6x + 4y = 9$
E $4x - 6y = 9$

- 4 The perpendicular bisector of the line segment joining $(1, 5)$ and $(7, 1)$ has equation
- A $3x - 2y = 6$
 - B $2x - 3y = -1$
 - C $3x + 2y = 18$
 - D $2x + 3y = 17$
 - E $y = \frac{3}{2}x - 3$
 - F $3x - 2y = 18$
- 5 The line $ax + by = 12$ has x -intercept 3 and y -intercept -4 . Find $a - b$.
- A 12
 - B 7
 - C -1
 - D 0
 - E -7
 - F 1
- 6 A line passes through $(1, 2)$ with gradient m . It crosses the x -axis at A and the y -axis at B , and the midpoint of AB is $(1, 2)$. Find m .
- A -2
 - B $\frac{1}{2}$
 - C 2
 - D -4
 - E $-\frac{1}{2}$
 - F -1

- 7 Two distinct lines have equations $y = m_1x + c_1$ and $y = m_2x + c_2$. Which of the following statements is/are true?

- I If $m_1m_2 = -1$ then the lines meet at exactly one point.
- II If $m_1 = m_2$ then the lines do not meet.
- III If the lines meet on the y -axis then $c_1 = c_2$.

- A none of them
- B I only
- C II only
- D III only
- E I and II only
- F I and III only
- G II and III only
- H I, II and III

- 8 The line L passes through $(4, 0)$ and $(0, 6)$. Find the perpendicular distance from the origin to L .

- A 5
- B $\frac{24}{13}$
- C $\frac{13}{12}$
- D $\frac{\sqrt{13}}{2}$
- E $2\sqrt{13}$
- F $\sqrt{13}$

- 9 The lines $y = 2x + 1$, $y = -x + 7$ and the x -axis enclose a triangle. Find its area.

- A $\frac{25}{2}$
- B $\frac{75}{4}$
- C $\frac{15}{2}$
- D $\frac{35}{2}$
- E 15
- F $\frac{45}{4}$

- 10** For which value of k are the lines $kx + 3y = 2$ and $2x - (k - 1)y = 5$ perpendicular?
- A** 3
B -1
C $\frac{3}{2}$
D 6
E -3
F 1
- 11** The point P lies on the line $y = 2x - 1$ and is equidistant from $(0, 0)$ and $(4, 2)$. Find the coordinates of P .
- A** $(2, 1)$
B $\left(\frac{3}{2}, 2\right)$
C $(2, 3)$
D $\left(\frac{5}{4}, \frac{3}{2}\right)$
E $(3, 5)$
F $(1, 1)$
- 12** The vertices of a triangle are $A(0, 0)$, $B(6, 0)$ and $C(2, 4)$. Find the equation of the line through C that bisects the area of the triangle.
- A** $y = x + 2$
B $y = -2x + 8$
C $x = 2$
D $y = 2x$
E $y = 4x - 4$
F $y = -4x + 12$